

CẤU TRÚC THƯ MỤC FLUTTER — FEATURES

dat_san_247_mobile · Flutter · Cubit / BLoC · Retrofit + Dio · json_serializable
Dựa trên: Prisma Schema v7 (58 models) + Mobile App Specification (57 màn hình)

I. Pattern Chuẩn — Dựa trên example/category_rut_gon

Mọi feature đều tuân theo cùng 1 pattern. Đây là cấu trúc "rút gọn" đúng nhất từ ví dụ của dự án:

Tầng	Folder	File	Vai trò
DATA	data/models/	xxx_model.dart + .g.dart	JSON ↔ Dart (json_serializable codegen)
DATA	data/repositories/	xxx_repository.dart	Abstract interface + Implementation: business logic
DATA	data/services/	xxx_service.dart + .g.dart	Retrofit/Dio API calls (codegen tạo HTTP methods)
PRESENTATION	presentation/cubit/	xxx_cubit.dart + xxx_state.dart	State management (flutter_bloc Cubit)
PRESENTATION	presentation/pages/	xxx_page.dart	1 màn hình = 1 page file
PRESENTATION	presentation/widgets/	xxx_widget.dart / xxx_card.dart	Reusable UI components riêng cho feature

Quy tắc quan trọng:

- Page gọi Cubit, KHÔNG gọi trực tiếp Service hay Repository
- Repository là abstract class — testable, mockable. Implementation inject Dio Service vào.
- Service (.g.dart) chỉ định nghĩa Retrofit @GET/@POST/@PATCH — không viết logic trong đây.
- State class dùng Freezed hoặc viết tay: Initial | Loading | Success(data) | Error(message)
- Models dùng @JsonSerializable() + @JsonKey() cho camelCase ↔ snake_case tự động.

II. Core Layer — lib/core/

Tầng hạ tầng dùng cho toàn app. Không thuộc feature nào.

lib/	
├── core/	// Infrastructure layer – dùng cho toàn app
│ ├── api/	
│ │ ├── api_client.dart	// Dio instance + interceptors (401 auto-refresh)
│ │ ├── api_endpoints.dart	// Tất cả URL constants (theo NestJS routes)
│ │ ├── api_exception.dart	// AppException, NetworkException, ServerException
│ │ └── api_response.dart	// BaseResponse<T>, PaginatedResponse<T>
│ ├── di/	
│ │ ├── injection.dart	// GetIt service locator setup
│ │ └── injection.g.dart	// ← generated (injectable)
│ └── router/	
│ ├── app_router.dart	// GoRouter: routes theo role
│ ├── app_router.g.dart	// ← generated (go_router_builder)
│ └── route_guard.dart	// Redirect theo platform role + venue_staff check

```
├── storage/
│   ├── secure_storage.dart // flutter_secure_storage: access_token, refresh_token
│   └── local_storage.dart // SharedPreferences: theme, lang, onboarding_done
├── theme/
│   ├── app_theme.dart
│   ├── app_colors.dart
│   └── app_text_styles.dart
├── utils/
│   ├── currency_utils.dart // formatVND(amount) → "150.000 đ"
│   ├── date_utils.dart // formatDate, formatTime, fromNow VN
│   ├── validators.dart // email, phone VN (0xx), required
│   ├── fcm_utils.dart // Firebase FCM token, save to user_devices
│   ├── qr_utils.dart // Generate/parse check_in_code QR
│   └── widgets/ // Shared UI atoms (không thuộc feature nào)
│       ├── app_button.dart
│       ├── app_text_field.dart
│       ├── app_loading.dart
│       ├── app_error_widget.dart
│       ├── app_empty_state.dart
│       ├── status_badge.dart // BookingStatus, PaymentStatus chips
│       ├── role_guard_widget.dart // Ẩn/hiện theo VenueStaffRole
│       └── confirm_bottom_sheet.dart
```

III. Features Layer — lib/features/

Toàn bộ 57 màn hình được tổ chức theo 4 nhóm: auth · customer · shared · owner · venue_staff

🟦 auth/splash 🟢 customer/ (22 màn) 🟠 shared/ 🟡 owner/ (13 màn) 🟤 venue_staff/ (12 màn)

```
├── features/
│   └── // — SHARED AUTH (dùng chung 3 roles) —
│       ├── auth/
│       │   ├── data/
│       │   │   ├── models/
│       │   │   │   ├── auth_model.dart // LoginRequest, LoginResponse, TokenModel
│       │   │   │   ├── auth_model.g.dart // ← generated
│       │   │   │   ├── otp_model.dart // OtpRequest, OtpVerifyRequest (OtpType enum)
│       │   │   │   └── otp_model.g.dart
│       │   │   └── repositories/
│       │   │       └── auth_repository.dart // abstract + impl: login, register, sendOtp,
│       │   │           verifyOtp
│       │   ├── services/
│       │   │   ├── auth_service.dart // POST /auth/login, /auth/register, /auth/refresh
│       │   │   └── auth_service.g.dart
│       │   └── presentation/
│       │       ├── cubit/
│       │       │   ├── auth_cubit.dart
│       │       │   └── auth_state.dart // AuthInitial, AuthLoading, AuthSuccess, AuthError
│       │       ├── pages/
│       │       │   ├── login_page.dart // A-S2: email/phone + password
│       │       │   ├── register_page.dart // A-S3: full_name, email, phone, password
│       │       │   ├── otp_page.dart // A-S4: 6-digit OTP input + timer
│       │       │   └── forgot_password_page.dart // A-S5: email/phone → OTP → new password
│       │       └── widgets/
│       │           └── auth_form_field.dart
```

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└─ social_login_button.dart

// — SPLASH & ONBOARDING —

└─ splash/
  └─ presentation/
    └─ cubit/
      └─ splash_cubit.dart // Check app_versions (force update), auth status
    └─ pages/
      └─ splash_page.dart // A-S1: logo + version check
      └─ onboarding_page.dart // Swipeable intro slides

// — CUSTOMER FEATURES (22 màn - C-01 đến C-22) —

└─ customer/
  └─ home/ // C-01: Home Feed
    └─ data/
      └─ models/
        └─ banner_model.dart // banners + banner_pages
        └─ banner_model.g.dart
        └─ sport_type_model.dart // sport_types
        └─ sport_type_model.g.dart
      └─ repositories/
        └─ home_repository.dart // getFeaturedVenues, getBanners, getSportTypes
      └─ services/
        └─ home_service.dart // GET /home/banners, /sport-types
        └─ home_service.g.dart
    └─ presentation/
      └─ cubit/
        └─ home_cubit.dart
        └─ home_state.dart
      └─ pages/
        └─ home_page.dart
      └─ widgets/
        └─ banner_carousel.dart // banners.mobile_image_url, auto_slide,
slide_duration
        └─ sport_category_grid.dart
        └─ featured_venue_card.dart
        └─ promo_banner_widget.dart

    └─ venue_search/ // C-02: Tìm kiếm & Danh sách Venue
      └─ data/
        └─ models/
          └─ venue_model.dart // venues, amenities, sport_assignments, rating
          └─ venue_model.g.dart
          └─ venue_filter_model.dart // Filter params: sport, city, district, price
fields
range
      └─ repositories/
        └─ venue_repository.dart
      └─ services/
        └─ venue_service.dart // GET /venues?sport=&city=&district=
        └─ venue_service.g.dart
      └─ presentation/
        └─ cubit/
          └─ venue_search_cubit.dart
          └─ venue_search_state.dart
        └─ pages/
          └─ venue_search_page.dart
        └─ widgets/
          └─ venue_card.dart // thumbnail, rating, total_reviews, is_featured
badge
          └─ venue_filter_sheet.dart

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└─ search_bar_widget.dart // lưu search_history

└─ venue_detail/ // C-03: Chi tiết Venue

- └─ data/
 - └─ models/
 - venue_detail_model.dart // venue + courts + operating_hours + exceptions
 - venue_detail_model.g.dart
 - court_model.dart // courts + pricing_rules + amenities
 - court_model.g.dart
 - operating_hours_model.dart // venue_operating_hours (DayOfWeek)
 - operating_hours_model.g.dart
 - └─ repositories/
 - venue_detail_repository.dart
 - └─ services/
 - venue_detail_service.dart // GET /venues/{slug}, /venues/{id}/courts
 - venue_detail_service.g.dart
- └─ presentation/
 - └─ cubit/
 - venue_detail_cubit.dart
 - venue_detail_state.dart
 - └─ pages/
 - venue_detail_page.dart
 - └─ widgets/
 - venue_gallery.dart // media_attachments, is_cover
 - venue_info_card.dart
 - operating_hours_widget.dart
 - amenities_grid.dart
 - court_list_tile.dart
 - review_summary_widget.dart // rating, rating_cleanliness, facilities, staff

└─ booking/ // C-05 đến C-08: Toàn bộ booking flow

- └─ data/
 - └─ models/
 - booking_model.dart // bookings + addons + status_history
 - booking_model.g.dart
 - time_slot_model.dart // Slot availability từ bookings + pricing_rules
 - time_slot_model.g.dart
 - booking_addon_model.dart // booking_addons + venue_services
 - booking_addon_model.g.dart
 - create_booking_request.dart // BookingDate, startTime, endTime, courtId
 - create_booking_request.g.dart
 - └─ repositories/
 - booking_repository.dart
 - └─ services/
 - booking_service.dart // POST /bookings, GET /courts/{id}/availability
 - booking_service.g.dart
- └─ presentation/
 - └─ cubit/
 - time_slot_cubit.dart // C-05: fetch slots, check conflicts, calc price
 - time_slot_state.dart
 - booking_cubit.dart // C-06,07,08: confirm flow
 - booking_state.dart
 - └─ pages/
 - time_slot_page.dart // C-05: calendar + slot grid + pricing
 - booking_confirm_page.dart // C-06: addon + voucher + total
 - payment_page.dart // C-07: PaymentMethod selector
 - booking_success_page.dart // C-08: QR + invoice
 - └─ widgets/
 - slot_grid_widget.dart // Time slots 30-min intervals
 - pricing_chip.dart // Hiển thị giá từ pricing_rules
 - addon_selector.dart // venue_services picker
 - promo_code_input.dart // promotions + user_vouchers
 - payment_method_selector.dart // PaymentMethod enum

```

├── order_summary_card.dart
├── check_in_qr_widget.dart // bookings.check_in_code QR
├── my_bookings/ // C-09, C-10, C-11: Quản lý booking
│   ├── data/
│   │   ├── models/
│   │   │   ├── my_booking_model.dart // bookings (list view)
│   │   │   ├── my_booking_model.g.dart
│   │   │   ├── cancel_booking_model.dart // cancellation_reason, refund calc
│   │   │   └── cancel_booking_model.g.dart
│   │   ├── repositories/
│   │   │   └── my_booking_repository.dart
│   │   ├── services/
│   │   │   ├── my_booking_service.dart // GET /bookings/me, PATCH /bookings/{id}/cancel
│   │   │   └── my_booking_service.g.dart
│   │   └── presentation/
│   │       ├── cubit/
│   │       │   ├── my_bookings_cubit.dart
│   │       │   ├── my_bookings_state.dart
│   │       │   ├── booking_detail_cubit.dart
│   │       │   └── booking_detail_state.dart
│   │       ├── pages/
│   │       │   ├── my_bookings_page.dart // C-09: list + status filter tabs
│   │       │   ├── booking_detail_page.dart // C-10: full detail + addons + history
│   │       │   └── cancel_booking_page.dart // C-11: refund calc + confirm
│   │       └── widgets/
│   │           ├── booking_list_tile.dart
│   │           ├── booking_status_timeline.dart // booking_status_history
│   │           ├── refund_info_card.dart // refund_policies + refund_rules
│   │           └── invoice_section.dart // invoices + invoice_items
│   └── review/ // C-12: Viết đánh giá
│       ├── data/
│       │   ├── models/
│       │   │   ├── review_model.dart // reviews (rating, cleanliness, facilities, staff)
│       │   │   └── review_model.g.dart
│       │   ├── repositories/
│       │   │   └── review_repository.dart
│       │   ├── services/
│       │   │   ├── review_service.dart // POST /reviews, GET /venues/{id}/reviews
│       │   │   └── review_service.g.dart
│       │   └── presentation/
│       │       ├── cubit/
│       │       │   ├── review_cubit.dart
│       │       │   └── review_state.dart
│       │       ├── pages/
│       │       │   └── write_review_page.dart
│       │       └── widgets/
│       │           ├── star_rating_input.dart // rating, cleanliness, facilities, staff
│       │           └── review_image_picker.dart // media_attachments (review_id)
│   └── recurring_booking/ // C-13: Đặt lịch định kỳ
│       ├── data/
│       │   ├── models/
│       │   │   ├── recurring_booking_model.dart // recurring_bookings + recurring_booking_days
│       │   │   └── recurring_booking_model.g.dart
│       │   ├── repositories/
│       │   │   └── recurring_booking_repository.dart
│       │   ├── services/
│       │   │   ├── recurring_booking_service.dart // GET/POST/PATCH /bookings/recurring
│       │   │   └── recurring_booking_service.g.dart
│       └── presentation/
│           └── cubit/

```

```

├── recurring_booking_cubit.dart
├── recurring_booking_state.dart
├── pages/
│   └── recurring_booking_page.dart
├── widgets/
│   └── recurring_schedule_card.dart // RecurringType, DayOfWeek chips
└── waitlist/ // C-14: Danh sách chờ
    ├── data/
    │   ├── models/
    │   │   ├── waitlist_model.dart // booking_waitlist (WaitlistStatus, priority)
    │   │   └── waitlist_model.g.dart
    │   ├── repositories/
    │   │   └── waitlist_repository.dart
    │   ├── services/
    │   │   ├── waitlist_service.dart // GET/DELETE /waitlist/me
    │   │   └── waitlist_service.g.dart
    │   └── presentation/
    │       ├── cubit/
    │       │   ├── waitlist_cubit.dart
    │       │   └── waitlist_state.dart
    │       ├── pages/
    │       │   └── waitlist_page.dart
    │       └── widgets/
    │           └── waitlist_item_tile.dart
    ├── wallet/ // C-15, C-16: Ví + Hóa đơn
    │   ├── data/
    │   │   ├── models/
    │   │   │   ├── wallet_model.dart // wallets (balance, locked_balance)
    │   │   │   ├── wallet_model.g.dart
    │   │   │   ├── transaction_model.dart // transactions (TransactionType, status)
    │   │   │   ├── transaction_model.g.dart
    │   │   │   ├── invoice_model.dart // invoices + invoice_items (InvoiceItemType)
    │   │   │   └── invoice_model.g.dart
    │   │   ├── repositories/
    │   │   │   └── wallet_repository.dart
    │   │   ├── services/
    │   │   │   ├── wallet_service.dart // GET /wallet/me, POST /wallet/deposit
    │   │   │   └── wallet_service.g.dart
    │   │   └── presentation/
    │   │       ├── cubit/
    │   │       │   ├── wallet_cubit.dart
    │   │       │   └── wallet_state.dart
    │   │       ├── pages/
    │   │       │   ├── wallet_page.dart // C-15: balance + transactions
    │   │       │   ├── deposit_page.dart
    │   │       │   └── invoice_page.dart // C-16: invoice list + detail + pdf
    │   │       └── widgets/
    │   │           ├── wallet_balance_card.dart
    │   │           ├── transaction_list_tile.dart
    │   │           └── invoice_card.dart
    │   └── promotion/ // C-17: Khuyến mãi & Voucher
    │       ├── data/
    │       │   ├── models/
    │       │   │   ├── promotion_model.dart // promotions (PromotionDiscountType, valid_from/to)
    │       │   │   ├── promotion_model.g.dart
    │       │   │   ├── voucher_model.dart // user_vouchers (VoucherStatus)
    │       │   │   └── voucher_model.g.dart
    │       │   ├── repositories/
    │       │   │   └── promotion_repository.dart
    │       │   └── services/

```



```

    └─ services/
      └─ profile_service.dart // GET/PATCH /users/me, /users/me/profile
      └─ profile_service.g.dart
    └─ presentation/
      └─ cubit/
        └─ profile_cubit.dart
        └─ profile_state.dart
      └─ pages/
        └─ profile_page.dart
        └─ edit_profile_page.dart
        └─ notification_settings_page.dart // user_profiles.notif_* fields
      └─ widgets/
        └─ sport_preference_chip.dart // user_sport_preferences
        └─ bank_account_tile.dart

```

// — SHARED FEATURES (Notification + Report - dùng chung 3 roles) —

```

└─ shared/
  └─ notification/ // C-19, VS-11: Thông báo
    └─ data/
      └─ models/
        └─ notification_model.dart // notifications (NotificationType, reference_type)
        └─ notification_model.g.dart
      └─ repositories/
        └─ notification_repository.dart
      └─ services/
        └─ notification_service.dart // GET /notifications, PATCH /notifications/read-
all      └─ notification_service.g.dart
    └─ presentation/
      └─ cubit/
        └─ notification_cubit.dart
        └─ notification_state.dart
      └─ pages/
        └─ notification_page.dart
      └─ widgets/
        └─ notification_tile.dart
        └─ notification_badge.dart // unread count badge
    └─ report/ // C-21: Báo cáo vi phạm (bottom sheet)
      └─ data/
        └─ models/
          └─ report_model.dart // reports (ReportTargetType, ReportReason)
          └─ report_model.g.dart
        └─ repositories/
          └─ report_repository.dart
        └─ services/
          └─ report_service.dart // POST /reports
          └─ report_service.g.dart
      └─ presentation/
        └─ cubit/
          └─ report_cubit.dart
          └─ report_state.dart
        └─ widgets/
          └─ report_bottom_sheet.dart // ReportReason selector

```

// — OWNER FEATURES (13 màn - 0-01 đến 0-13) —

```

└─ owner/
  └─ dashboard/ // 0-01
    └─ data/
      └─ models/
        └─ owner_dashboard_model.dart // booking stats + commission summary
        └─ owner_dashboard_model.g.dart

```



```

    repositories/
    | owner_dashboard_repository.dart
    |
    | services/
    | | owner_dashboard_service.dart // GET /owner/dashboard/stats
    | | owner_dashboard_service.g.dart
    |
    | presentation/
    | | cubit/
    | | | owner_dashboard_cubit.dart
    | | | owner_dashboard_state.dart
    | |
    | | pages/
    | | | owner_dashboard_page.dart
    | |
    | | widgets/
    | | | stat_summary_card.dart
    | | | recent_bookings_widget.dart
    |
    | venue_management/ // 0-02, 0-03: Venue CRUD
    | | data/
    | | | models/
    | | | | owner_venue_model.dart // venues (owner view + operating_hours +
exceptions) | | | | owner_venue_model.g.dart
    | | | | update_venue_request.dart
    | | | | update_venue_request.g.dart
    | | |
    | | | repositories/
    | | | | venue_management_repository.dart
    | | |
    | | | services/
    | | | | venue_management_service.dart // PATCH /owner/venues/{id}, /operating-hours
    | | | | venue_management_service.g.dart
    | | |
    | | | presentation/
    | | | | cubit/
    | | | | | venue_management_cubit.dart
    | | | | | venue_management_state.dart
    | | | |
    | | | | pages/
    | | | | | my_venues_page.dart
    | | | | | edit_venue_page.dart
    | | | |
    | | | | widgets/
    | | | | | operating_hours_editor.dart // venue_operating_hours per DayOfWeek
    | | | | | schedule_exception_tile.dart // venue_schedule_exceptions
    | | | | | social_links_form.dart
    |
    | court_management/ // 0-04: Courts CRUD
    | | data/
    | | | models/
    | | | | court_manage_model.dart // courts (CourtSurfaceType, display_order)
    | | | | court_manage_model.g.dart
    | | |
    | | | repositories/
    | | | | court_management_repository.dart
    | | |
    | | | services/
    | | | | court_management_service.dart // CRUD /owner/venues/{id}/courts
    | | | | court_management_service.g.dart
    | | |
    | | | presentation/
    | | | | cubit/
    | | | | | court_management_cubit.dart
    | | | | | court_management_state.dart
    | | | |
    | | | | pages/
    | | | | | courts_list_page.dart
    | | | | | edit_court_page.dart
    | | | |
    | | | | widgets/
    | | | | | court_manage_card.dart
    |
    | pricing/ // 0-05: Bảng giá
    | | data/
    | | | models/

```



```

├── owner_booking_service.dart // GET /owner/bookings, PATCH confirm/cancel
├── owner_booking_service.g.dart
├── presentation/
│   ├── cubit/
│   │   ├── booking_calendar_cubit.dart
│   │   └── booking_calendar_state.dart
│   ├── pages/
│   │   ├── booking_calendar_page.dart // 0-08: calendar grid by court
│   │   └── owner_booking_detail_page.dart // 0-09
│   └── widgets/
│       ├── calendar_grid_widget.dart
│       └── booking_slot_chip.dart
├── staff_management/ // 0-10
│   ├── data/
│   │   ├── models/
│   │   │   ├── venue_staff_model.dart // venue_staff (VenueStaffRole, work_days)
│   │   │   ├── venue_staff_model.g.dart
│   │   │   ├── staff_invite_model.dart // venue_staff_invites
│   │   │   └── staff_invite_model.g.dart
│   │   ├── repositories/
│   │   │   └── staff_management_repository.dart
│   │   ├── services/
│   │   │   ├── staff_management_service.dart // GET/POST/PATCH /venue-staff, /invites
│   │   │   └── staff_management_service.g.dart
│   └── presentation/
│       ├── cubit/
│       │   ├── staff_management_cubit.dart
│       │   └── staff_management_state.dart
│       ├── pages/
│       │   ├── staff_list_page.dart
│       │   └── invite_staff_page.dart
│       └── widgets/
│           └── staff_member_tile.dart // VenueStaffRole badge, work_days
├── revenue/ // 0-11
│   ├── data/
│   │   ├── models/
│   │   │   ├── commission_model.dart // commission_records (CommissionStatus)
│   │   │   ├── commission_model.g.dart
│   │   │   ├── payout_request_model.dart // payout_requests (PayoutStatus)
│   │   │   └── payout_request_model.g.dart
│   │   ├── repositories/
│   │   │   └── revenue_repository.dart
│   │   ├── services/
│   │   │   ├── revenue_service.dart // GET /owner/revenue, POST /payout-requests
│   │   │   └── revenue_service.g.dart
│   └── presentation/
│       ├── cubit/
│       │   ├── revenue_cubit.dart
│       │   └── revenue_state.dart
│       ├── pages/
│       │   ├── revenue_page.dart
│       │   └── payout_request_page.dart
│       └── widgets/
│           ├── revenue_chart_widget.dart
│           └── commission_list_tile.dart
├── review_management/ // 0-12
│   ├── data/
│   │   ├── models/
│   │   │   ├── review_manage_model.dart
│   │   │   └── review_manage_model.g.dart

```

```

    repositories/
      review_management_repository.dart
    services/
      review_management_service.dart // GET /owner/reviews, POST
/reviews/{id}/respond
      review_management_service.g.dart
    presentation/
      cubit/
        review_management_cubit.dart
        review_management_state.dart
      pages/
        review_management_page.dart
      widgets/
        review_respond_card.dart // reviews.response field

  venue_verification/ // 0-13
    data/
      models/
        verification_model.dart // venue_verifications (VerificationStatus)
        verification_model.g.dart
      repositories/
        verification_repository.dart
      services/
        verification_service.dart // POST /venues/{id}/verifications
        verification_service.g.dart
    presentation/
      cubit/
        verification_cubit.dart
        verification_state.dart
      pages/
        venue_verification_page.dart
      widgets/
        document_upload_tile.dart // business_license, id_card_front/back

// — VENUE STAFF FEATURES (12 màn – VS-01 đến VS-12) —

  venue_staff/
    dashboard/ // VS-01
      data/
        models/
          staff_dashboard_model.dart // booking counts + court status summary
          staff_dashboard_model.g.dart
        repositories/
          staff_dashboard_repository.dart
        services/
          staff_dashboard_service.dart // GET /venue-staff/dashboard
          staff_dashboard_service.g.dart
      presentation/
        cubit/
          staff_dashboard_cubit.dart
          staff_dashboard_state.dart
        pages/
          staff_dashboard_page.dart
        widgets/
          count_status_row.dart
          today_stats_card.dart

    check_in/ // VS-02, VS-03: QR Scanner
      data/
        models/
          check_in_model.dart // bookings (check_in_code, CHECKED_IN)
          check_in_model.g.dart
      repositories/

```

```

├── check_in_repository.dart
├── services/
│   ├── check_in_service.dart // POST /venue-staff/check-in (check_in_code)
│   └── check_in_service.g.dart
├── presentation/
│   ├── cubit/
│   │   ├── check_in_cubit.dart
│   │   └── check_in_state.dart
│   ├── pages/
│   │   ├── qr_scanner_page.dart // VS-02: camera QR + manual input
│   │   └── check_in_confirm_page.dart // VS-03: customer info + addons
│   └── widgets/
│       ├── qr_scan_overlay.dart
│       └── check_in_result_card.dart
├── schedule/ // VS-04, VS-06: Lịch hôm nay + tuần
│   ├── data/
│   │   ├── models/
│   │   │   ├── staff_booking_model.dart // bookings (venue_id, date, status)
│   │   │   └── staff_booking_model.g.dart
│   │   ├── repositories/
│   │   │   └── schedule_repository.dart
│   │   ├── services/
│   │   │   ├── schedule_service.dart // GET /venue-staff/bookings?date=today
│   │   │   └── schedule_service.g.dart
│   └── presentation/
│       ├── cubit/
│       │   ├── schedule_cubit.dart
│       │   └── schedule_state.dart
│       ├── pages/
│       │   ├── today_schedule_page.dart // VS-04
│       │   ├── weekly_schedule_page.dart // VS-06
│       │   └── staff_booking_detail_page.dart // VS-05
│       └── widgets/
│           ├── timeline_slot_widget.dart
│           ├── staff_booking_tile.dart
│           └── addon_add_sheet.dart // VS-05: thêm addon (STAFF+MANAGER)
├── court_status/ // VS-07: Trạng thái sân
│   ├── data/
│   │   ├── models/
│   │   │   ├── court_status_model.dart // courts + bookings (realtime) + court_maintenance
│   │   │   └── court_status_model.g.dart
│   │   ├── repositories/
│   │   │   └── court_status_repository.dart
│   │   ├── services/
│   │   │   ├── court_status_service.dart
│   │   │   └── court_status_service.g.dart
│   └── presentation/
│       ├── cubit/
│       │   ├── court_status_cubit.dart
│       │   └── court_status_state.dart
│       ├── pages/
│       │   └── court_status_page.dart
│       └── widgets/
│           ├── court_status_card.dart // AVAILABLE / IN_USE / MAINTENANCE
│           └── maintenance_form.dart // court_maintenance (MANAGER only)
├── staff_profile/ // VS-12: Hồ sơ nhân viên
│   ├── data/
│   │   ├── models/
│   │   │   ├── staff_profile_model.dart // venue_staff + venues info
│   │   │   └── staff_profile_model.g.dart

```



IV. Bảng Tổng hợp — Feature → DB Tables → Service

Map từng feature folder → DB tables liên quan → service file → màn hình:

Feature Folder	Màn hình	DB Tables chính	Service file	Pages / Widgets chính
auth	Shared Auth	users, otp_verifications, refresh_tokens, user_devices	auth_service, otp_model	5 màn: login, register, otp, forgot, reset
splash	Splash	app_versions	splash_cubit	2 màn: splash, onboarding
customer/home	C-01 Home	banners, venues (featured), sport_types, promotions	home_service	banner_carousel, sport_category_grid
customer/venue_search	C-02 Tìm sân	venues, sport_assignments, amenities, search_history	venue_service	venue_card, filter_sheet
customer/venue_detail	C-03,04 Venue Detail	venues, courts, reviews, media_attachments, operating_hours	venue_detail_service	gallery, operating_hours, review_summary
customer/booking	C-05~08 Booking Flow	bookings, pricing_rules, payments, transactions, invoices, promotions	booking_service	4 pages: slot→confirm→pay→success
customer/my_bookings	C-09~11 My Bookings	bookings, booking_status_history, refund_policies, refund_rules	my_booking_service	list + detail + cancel
customer/review	C-12 Review	reviews, media_attachments (review_id)	review_service	star_rating_input, image_picker
customer/recurring_booking	C-13 Recurring	recurring_bookings, recurring_booking_days	recurring_booking_service	RecurringType, DayOfWeek
customer/waitlist	C-14 Waitlist	booking_waitlist (WaitlistStatus)	waitlist_service	waitlist_item_tile
customer/wallet	C-15,16 Wallet	wallets, transactions, invoices, invoice_items	wallet_service	balance, history, invoice
customer/promotion	C-17 Promotions	promotions, user_vouchers (VoucherStatus)	promotion_service	promo_card, voucher_card
customer/favorite	C-18 Favorites	favorite_venues	favorite_service	favorite_venue_card

Feature Folder	Màn hình	DB Tables chính	Service file	Pages / Widgets chính
shared/notification	C-19, VS-11	notifications (NotificationType, reference_type)	notification_service	notification_tile, badge
customer/support	C-20 Support	support_tickets, support_ticket_messages	support_service	ticket list + chat
shared/report	C-21 Report	reports (ReportTargetType, ReportReason)	report_service	report_bottom_sheet
customer/profile	C-22 Profile	users, user_profiles, user_sport_preferences, payout_bank_accounts	profile_service	sport pref + notif settings
owner/dashboard	O-01 Dashboard	bookings (stats), commission_records, reviews	owner_dashboard_service	stat_summary, recent_bookings
owner/venue_management	O-02,03 Venue Mgmt	venues, venue_operating_hours, venue_schedule_exceptions	venue_management_service	hours_editor, exception_tile
owner/court_management	O-04 Courts	courts, sport_assignments, amenities, media_attachments	court_management_service	court_manage_card
owner/pricing	O-05 Pricing	pricing_rules (DayOfWeek, time, priority)	pricing_service	pricing_rule_tile
owner/services_management	O-06 Services	venue_services (ServiceUnit, VenueServiceType)	venue_service_api	service_item_tile
owner/refund_policy	O-07 Refund	refund_policies, refund_rules	refund_policy_service	refund_rule_tile
owner/booking_calendar	O-08,09 Calendar	bookings, booking_waitlist, booking_status_history	owner_booking_service	calendar_grid, slot_chip
owner/staff_management	O-10 Staff	venue_staff, venue_staff_invites (VenueStaffRole)	staff_management_service	staff_tile, invite_form
owner/revenue	O-11 Revenue	commission_records, payout_requests, wallets	revenue_service	chart, commission_tile
owner/review_management	O-12 Reviews	reviews (response, responded_by)	review_management_service	review_respond_card
owner/venue_verification	O-13 Verify	venue_verifications (VerificationStatus)	verification_service	document_upload_tile
venue_staff/dashboard	VS-01 Dashboard	bookings (today), courts, court_maintenance	staff_dashboard_service	court_status_row
venue_staff/check_in	VS-02,03 Check-in	bookings (check_in_code → CHECKED_IN)	check_in_service	QR scanner + confirm
venue_staff/schedule	VS-04~06 Schedule	bookings, court_maintenance	schedule_service	timeline, weekly grid
venue_staff/court_status	VS-07 Courts	courts, bookings (current), court_maintenance	court_status_service	status card MANAGER+
venue_staff/staff_profile	VS-12 Profile	venue_staff (role, work_days), venues	staff_profile_service	venue_role_badge

V. Dart Enums — Mirror từ Prisma Schema

Tạo file lib/core/utils/app_enums.dart mirror các enum từ DB để type-safe khi parse JSON:

```
// lib/core/utils/app_enums.dart
// Mirror các Prisma enums quan trọng nhất

enum BookingStatus { pending, confirmed, checkedIn, completed, cancelled, noShow }
enum PaymentMethod { cash, vnpay, momo, zalopay, bankTransfer, wallet }
enum PaymentStatus { pending, paid, failed, refunded, partiallyRefunded }
enum VenueStaffRole { owner, manager, staff, receptionist }
enum TransactionType { deposit, payment, refund, payout, commission, adjustment }
enum WaitlistStatus { waiting, converted, expired, cancelled }
enum RecurringType { daily, weekly, biweekly, monthly }
enum NotificationType { bookingConfirmed, bookingCancelled, bookingReminder,
    paymentSuccess, paymentFailed, reviewReceived,
    promotion, payoutProcessed, venueStaffInvite, system }
enum VoucherStatus { unused, used, expired }
enum ReportTargetType { user, venue, review, transaction }
enum ReportReason { spam, inappropriate, fake, fraud, disputeTransaction, other }
enum VenueStatus { pending, approved, rejected, suspended }
enum CommissionStatus { pending, approved, paid, cancelled }
enum PayoutStatus { pending, processing, completed, rejected, cancelled }
enum InvoiceItemType { court, addon, vat, discount }
enum ServiceUnit { unit, hour, session, person, set_ }
enum DayOfWeek { monday, tuesday, wednesday, thursday, friday, saturday, sunday }

// Dùng với json_serializable:
// @JsonKey(unknownEnumValue: BookingStatus.pending)
// BookingStatus status;
```

VI. Dependency Injection — GetIt + Injectable

```
// lib/core/di/injection.dart

@InjectableInit()
void configureDependencies() => getIt.init();

// Mỗi Repository được đăng ký như sau:
@Injectable(as: BookingRepository)
class BookingRepositoryImpl implements BookingRepository {
    final BookingService _service;
    BookingRepositoryImpl(this._service);
}

// Mỗi Service (Retrofit):
@lazySingleton
BookingService bookingService(Dio dio) => BookingService(dio);

// Mỗi Cubit inject Repository:
@injectable
class BookingCubit extends Cubit<BookingState> {
    final BookingRepository _repo;
    BookingCubit(this._repo) : super(const BookingState.initial());
}
```

VII. Thống kê

Metric	Số lượng	Ghi chú
Feature folders	33	auth, splash + 10 customer + 2 shared + 13 owner + 8 venue_staff
Total pages (.dart)	~57	Khớp đúng với Mobile App Specification
Service files	33 + .g.dart	1 service per feature, codegen tạo HTTP client
Repository files	33	Abstract interface để mock trong unit test
Cubit + State	$33 \times 2 = 66$	1 cubit + 1 state file per feature
Model files	~70 + .g.dart	Mirror 58 Prisma models → Dart classes
Dart Enums	18 enums	Mirror toàn bộ Prisma enums quan trọng
Core widgets	~10	Shared atoms: button, loading, error, status_badge, role_guard
DB tables covered	48 / 58	10 bảng hệ thống (audit_logs, settings...) không cần trong mobile

Ghi chú: venue_staff/pricing/ và venue_staff/staff_management/ (VS-08, VS-09) tái dụng model từ owner/pricing và owner/staff_management — không cần duplicate data layer, chỉ cần presentation layer riêng.